



Sandya Mannarswamy

Welcome to this month's CodeSport, in which we continue our discussion on multi-threaded code and discuss lock granularity.

Last month's takeaway problem came from one of our readers, Nilesh Govande. Given a two-dimensional $n \times n$ matrix M of positive and negative integers, readers were to find the sub-rectangle with the largest sum. The sum of a rectangle is the sum of all the elements in that rectangle.

In this problem, the sub-rectangle with the largest sum is referred to as the *maximal sub-rectangle*. A sub-rectangle is any contiguous sub-array of size 1×1 or greater, located within the whole $n \times n$ array.

Three of our readers had sent their answers to this problem. However, none of them were correct. Hence I am going to keep this problem open for this month also. However, let me provide some clues.

The problem of a maximal sub-rectangle is based on a similar variant of the problem, namely that of calculating the maximum contiguous sub-array in one dimension.

That problem can be solved in $O(n)$ time (time linear to the size of the input array). We had discussed this problem in an earlier column. Therefore, extending that idea, the first approach would be to compute the sums of all possible rectangular sub-arrays $A[i_1-i_2][j_1-j_2]$

where $0 < i_1, i_2 < n$ and $0 < j_1, j_2 < n$.

Let $S[r,c]$ denote the sum of the rectangular sub-array of M with one corner at entry $[0,0]$ and the other at $[r, c]$. This can be computed in $O(n^2)$ time.

Observe that the sum of any rectangular sub-array of M can be computed in constant time given the table S . Once we have the table S , we can just guess different values for the lower-left and the upper-right corner of the rectangular sub-array and use the S table to compute its sum. The sub-array that yields the maximum sum is our result. Such an algorithm will be of the order of $O(n^4)$.

I leave it to the readers to write down the actual code to compute this. Can you come up with a $O(n^3)$ algorithm? Please do send in your solutions.

Now we will continue our discussion on multi-threaded code, while looking at locks and the issue of lock contention.

Mutual exclusion locks

As we all know, a binary semaphore or a mutex, as it is more popularly known, is the simplest type of lock. Though there are other types of locks such as adaptive locks, read-write locks, queue-based locks, etc, for our discussion on lock contention, we will focus on the mutex.